

I built a customer  
risk model.  
Then I tested it on  
the customers who  
actually caused UK  
banks to be fined.

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Excel and Power Query. No machine learning, on purpose.

## THE IDEA

# Most risk models are judged on whether the maths works.

That is the easy test. A customer risk assessment can be perfectly consistent and still rate the wrong people Low.

So I built one properly — 20 risk factors mapped to regulation 18(2)(b) of the Money Laundering Regulations, 98 level definitions, every weight carrying a written rationale and a published source — and then rebuilt six customers named in FCA enforcement notices as they looked on the day each bank took them on.

**Each one is scored twice: once reading every open question in the customer's favour, once at the worst reading the notice does not rule out.**

Because a reconstruction from a published notice is twenty judgement calls, and reporting one number hides nineteen of them.

THE RESULT, AND THE PROBLEM WITH IT

# Not one reaches High on the arithmetic.

In any reading. One of the six is rated High, and only because a mandatory escalator fires.

# 3.37

is the highest score any of them *could* have reached, with every open factor pushed to 5. High begins at 3.50.

The facts each notice settles — a UK company, a UK address, branch onboarding — pin 41–69% of the model's weight at level 1 before any judgement is made. The headline finding was fixed by the inputs. That is not a discovery, and I only know it because someone checked.

FINDING ONE

# One word decided the most serious case.

Fowler Oldfield was a Bradford jewellery business. Around £365m was deposited with NatWest, roughly £264m of it in cash. It produced the FCA's first criminal prosecution under the money laundering regulations.

|                                       |                          |
|---------------------------------------|--------------------------|
| Read as a <b>cash-intensive trade</b> | <b>1.82</b> → <b>Low</b> |
|---------------------------------------|--------------------------|

|                                             |                           |
|---------------------------------------------|---------------------------|
| Read as a <b>dealer in high-value goods</b> | <b>1.90</b> → <b>High</b> |
|---------------------------------------------|---------------------------|

Same file, same day. The scores differ by 0.075 and neither crosses a band. The rating differs completely, because one reading fires an escalator and the other does not.

That is a defect in my level definitions, not in the analyst reading them. So I wrote a fix.

# My fix would have broken the only case the model got right.

Barclays opened an account for a gold refining and trading company in January 2015. The application form said £500,000 turnover, manually amended to £3 million, and that it would not trade outside the EU. Eleven days later the file recorded gold from Ghana and Burkina Faso.

**My model rates it High on day one** — 2.14, escalated — because "gold refining and trading" is a dealer in high-value goods under the same ambiguous definition that page four calls a defect.

Tie that level to *registration*, as I had recommended, and this customer drops to Medium. A gold trader declaring no cash would not have been registered.

I wrote a fix, tested it on the case that motivated it, and never asked what else it touched. The recommendation is still on the list, now with the argument against it attached.

### FINDING THREE

## A customer whose address was a London landmark scores 1.52.

Monzo onboarded customers who gave well-known London landmarks as their home address. My library has a factor written for exactly that — from that notice, before any of it was scored.

I then gave the factor **1.5% of the model**, and wrote a paragraph defending the weight.

Scored through, the customer lands comfortably in Low. The factor that saw the problem was too light to change the answer.

Both the reasoning and the result stay in the repository. Deleting the argument I got wrong would have removed the only evidence that the test worked.

FINDING FOUR

# The weights were never the thing that mattered.

I spent a whole document arguing about whether geography should carry 25% or 15%.

|                                                                                    |                                                  |
|------------------------------------------------------------------------------------|--------------------------------------------------|
| Move any category weight by 10 percentage points                                   | <b>up to 26 of 400<br/>customers change band</b> |
| The same change, against band boundaries placed where the population actually sits | <b>29 to 100<br/>customers change band</b>       |

The boundaries decide the output. The weights barely touch it.

Which reverses the order of the work: fix the bands first, argue about the weights second. Those boundaries are fitted to a population I invented, which is circular, and the recommendation says so where it is made rather than in a footnote.

# The 2026 EDD change removes the only control catching a whole population.

SI 2026/621, made 9 June 2026, narrows mandatory jurisdiction-based enhanced due diligence to the three FATF *Call for Action* countries, rather than those plus the 22 under *Increased Monitoring*.

|                                                 |          |
|-------------------------------------------------|----------|
| Had jurisdiction-based EDD before 2026          | 16       |
| Have it under the 2026 rule                     | 5        |
| Lose it                                         | 11       |
| <b>Of those, escalated for any other reason</b> | <b>0</b> |

The model still scores them. It just cannot move them across a band. The score sees the risk; the rating does not.



## 41 findings. Most of them were right.

I gave the finished pack to a system that had not seen it, told it to assume the work contained errors, and set three calibration questions first so I could tell reading from guessing. It answered all three.

- It found a reconstruction I had built from a narrative instead of from the Final Notice — page five is the rebuilt version, and it inverts what I had published
- It found that page three's headline was guaranteed by its inputs
- It found a regulation I had cited from a law firm's commentary rather than from the instrument
- Three findings were wrong, and are answered rather than accepted

All 41 dispositions are in the repository, including the three I dispute and the working behind them.

**Synthetic data, a fictional bank, and a self-directed learning project alongside a BA in Accounting and Finance. Not professional advice, and not independently validated by a person.** Full method, workbooks and case files on GitHub — [link in the comments](#).